© 2021 Cengage Learning, Cengage Learning

Chapter Review

Problems and Applications

1. Maya divides her income between coffee and croissants (both of which are normal goods). An early frost in Brazil causes a large increase in the price of coffee in the United States.

    a. Show the effect of the frost on Maya's budget constraint.

    b. Show the effect of the frost on Maya's optimal consumption bundle assuming that the substitution effect outweighs the income effect for croissants.

    c. Show the effect of the frost on Maya's optimal consumption bundle assuming that the income effect outweighs the substitution effect for croissants.

2. Compare the following two pairs of goods:

    - Coke and Pepsi
    - Skis and ski bindings

    a. In which case are the two goods complements? In which case are they substitutes?

    b. In which case do you expect the indifference curves to be fairly straight? In which case do you expect the indifference curves to be very bowed?

    c. In which case will the consumer respond more to a change in the relative price of the two goods?

3. You consume only soda and pizza. One day, the price of soda goes up, the price of pizza goes down, and you are just as happy as you were before the price changes.

    a. Illustrate this situation on a graph.

b. How does your consumption of the two goods change? How does your response depend on income and substitution effects?

c. Can you afford the bundle of soda and pizza you consumed before the price changes?

4. Raj consumes only cheese and crackers.

   a. Could cheese and crackers both be inferior goods for Raj? Explain.

   b. Suppose that cheese is a normal good for Raj while crackers are an inferior good. If the price of cheese falls, what happens to Raj's consumption of crackers? What happens to his consumption of cheese? Explain.

5. Darius buys only milk and cookies.

   a. In year 1, Darius earns $\$100$, milk costs $\$2$ per quart, and cookies cost $\$4$ per dozen. Draw Darius's budget constraint.

   b. Now suppose that all prices increase by 10 percent in year 2 and that Darius's salary increases by 10 percent as well. Draw Darius's new budget constraint. How would Darius's optimal combination of milk and cookies in year 2 compare to his optimal combination in year 1?

6. State whether each of the following statements is true or false. Explain your answers.

   a. "All Giffen goods are inferior goods."

   b. "All inferior goods are Giffen goods."

7. A college student has two options for meals: eating at the dining hall for $\$6$ per meal, or eating a Cup O' Soup for $\$1.50$ per meal. Her weekly food budget is $\$60$.

   a. Draw the budget constraint showing the trade-off between dining-hall meals and Cups O' Soup. Assuming that she spends equal amounts on both goods, draw an indifference curve showing the optimum choice. Label the optimum as point A.

   b. Suppose the price of a Cup O' Soup now rises to $\$2$. Using your diagram from part (a), show the consequences of this change in price. Assume that our student now spends only 30 percent of her income on dining-hall meals. Label the new optimum as point B.

c. What happened to the quantity of Cups O' Soup consumed as a result of this price change? What does this result say about the income and substitution effects? Explain.

d. Use points A and B to draw a demand curve for Cup O' Soup. What is this type of good called?

8. Consider your decision about how many hours to work.

   a. Draw your budget constraint assuming that you pay no taxes on your income. On the same diagram, draw another budget constraint assuming that you pay a 15 percent income tax.

   b. Show how the tax might lead you to work more hours, fewer hours, or the same number of hours. Explain.

9. Anya is awake for 100 hours per week. Using one diagram, show Anya's budget constraints if she earns $\$12$ per hour, $\$16$ per hour, and $\$20$ per hour. Now draw indifference curves such that Anya's labor-supply curve is upward-sloping when the wage is between $\$12$ and $\$16$ per hour and backward-sloping when the wage is between $\$16$ and $\$20$ per hour.

10. Draw the indifference curve for someone deciding how to allocate time between work and leisure. Suppose the wage increases. Is it possible that the person's consumption would fall? Is this plausible? Discuss. (*Hint*: Think about income and substitution effects.)

11. Economist George Stigler once wrote that, according to consumer theory, "if consumers do not buy less of a commodity when their incomes rise, they will surely buy less when the price of the commodity rises." Explain this statement using the concepts of income and substitution effects.

12. Five consumers have the following marginal utility of apples and pears:

	Marginal Utility of Apples	**Marginal Utility of Pears**
Claire	6	12
Phil	6	6

	Marginal Utility of Apples	Marginal Utility of Pears
Haley	6	3
Alex	3	6
Luke	3	12

The price of an apple is $\$1$, and the price of a pear is $\$2$. Which, if any, of these consumers are optimizing their choices of fruit? For those who are not, how should they change their spending?

© 2021 Cengage Learning, Cengage Learning

© 2026 Cengage Learning Inc. All rights reserved. No part of this work may by reproduced or used in any form or by any means - graphic, electronic, or mechanical, or in any other manner - without the written permission of the copyright holder.